Geoff Twardokus[1] and Hanif Rahbari[1]

[1]ESL Global Cybersecurity Institute, Rochester Institute of Technology

September 10, 2025

Abstract

The safety and reliability of next-generation connected vehicles (CVs) will depend on verifying the authenticity of vehicle-to-vehicle (V2V) safety messages. Today, these safety messages are secured using elliptic-curve digital signatures that are vulnerable to attacks using a quantum computer, and a transition to quantum-resistant cryptography (QRC) is imperative to guard against (near-)future quantum adversaries. However, handling the overhead of QRC (i.e., large keys and signatures) is non-trivial due to the bandwidth, latency, and message size constraints of 4G and 5G Cellular Vehicle-to-Everything (C-V2X) communication protocols, as well as backward compatibility requirements. Moreover, relatively immature QRC must be integrated alongside elliptic-curve algorithms in a crypto-agile manner to avoid reliance on any single algorithm. In this work, we present CHASM, a hybrid authentication protocol that brings quantum-resistant message authentication to C-V2X but preserves all existing (classical) security guarantees—a backward-compatible, crypto-agile solution. To accommodate QRC overhead while maintaining reliability of V2V safety services, CHASM combines careful certificate fragmentation with a novel adaptive modulation scheme—a cross-layer approach—that allows for efficient distribution of large hybrid certificates under tight latency bounds. We formally prove CHASM achieves EUF-CMA unforgeability against both classical and quantum adversaries, and we demonstrate its practicality through over the-air experiments using software-defined radios, supplemented by simulations where necessary for scalability. Our results showcase the negligible impact of CHASM on both communication reliability and safety message verification delay.

Posted on 10 Sep 2025 — CC-BY-NC-SA 4 — https://doi.org/10.36227/techrxiv.175751197.77663474/v1 — e-Prints posted on TechRxiv are preliminary reports that are not peer reviewed. They should not b...

Quantum-Resistant Safety Message Authentication for NextG C-V2X: A Cross-Layer Approach

Geoff Twardokus, *Member, IEEE,* and Hanif Rahbari, *Member, IEEE*

***Abstract*—The safety and reliability of next-generation connected vehicles (CVs) will depend on verifying the authenticity of vehicle-to-vehicle (V2V) safety messages. Today, these safety messages are secured using elliptic-curve digital signatures that are vulnerable to attacks using a quantum computer, and a transition to quantum-resistant cryptography (QRC) is imperative to guard against (near-)future quantum adversaries. However, handling the overhead of QRC (i.e., large keys and signatures) is non-trivial due to the bandwidth, latency, and message size constraints of 4G and 5G Cellular Vehicle-to-Everything (C-V2X) communication protocols, as well as backward compatibility requirements. Moreover, relatively immature QRC must be integrated alongside elliptic-curve algorithms in a crypto-agile manner to avoid reliance on any single algorithm. In this work, we present CHASM, a hybrid authentication protocol that brings quantum-resistant message authentication to C-V2X but preserves all existing (classical) security guarantees—a backward-compatible, crypto-agile solution. To accommodate QRC overhead while maintaining reliability of V2V safety services, CHASM combines careful certificate fragmentation with a novel adaptive modulation scheme—a cross-layer approach—that allows for efficient distribution of large hybrid certificates under tight latency bounds. We formally prove CHASM achieves EUF-CMA unforgeability against both classical and quantum adversaries, and we demonstrate its practicality through over-the-air experiments using software-defined radios, supplemented by simulations where necessary for scalability. Our results showcase the negligible impact of CHASM on both communication reliability and safety message verification delay.**

I. Introduction

Enhancing safety is a top priority in smart transportation systems. In 2024 alone, nearly 40,000 people in the U.S. [1] and 20,000 in the European Union [2] lost their lives in roadway crashes. To bring these fatalities to near zero, connected vehicle (CV) technology will play a critical role in advanced driver assistance systems (ADAS) by enabling vehicles to exchange safety messages for split-second decision-making informed by broader context—e.g., whether a driver should brake or swerve to avoid an *out-of-sight* hazard. However, as the authenticity of these messages may be compromised within 10–15 years by the emergence of cryptographically relevant quantum computers [3], [4], injection of falsified messages could mislead human or autonomous drivers, manipulate traffic flow, and even engineer crashes, once again raising safety risks. Transitioning to *quantum-resistant cryptography* (QRC)[1] is underway in many systems, but no solution exists to handle QRC's overhead in constrained, low-latency CV protocols yet.

Modern CVs rely on 4G, 5G, or future 6G Cellular Vehicle-to-Everything (C-V2X) technology to directly exchange safety messages with low latency and high reliability [5]–[7]. C-V2X sidelink is designed for dynamic, fast-fading wireless channels and limited shared bandwidth in part by putting a tight cap on message size (just 867 bytes in a typical configuration[2]). Until now, this has not been a security problem because with small (32-byte) Elliptic Curve Digital Signature Algorithm (ECDSA) signatures employed to ensure message authentication and integrity, the total packet size is only 200–300 bytes [9]. However, as Fig. 1 shows, QRC algorithms standardized by the National Institute of Standards and Technology (NIST) require anywhere from $\sim 1,500$–$17,000$ bytes just for a public key and signature—without even including a safety message, headers, and metadata—making a safe, secure, and practical transition to QRC a daunting challenge in C-V2X.

Packet size restrictions also challenge adopting QRC in other constrained network protocols, such as IPsec [10] and wireless sensor networks [11], albeit with more relaxed caps. However, specific challenges in the *vehicular* domain make this transition uniquely complex for vehicle-to-vehicle (V2V) communications. Safety messages in V2V are broadcast and ephemeral by nature, meaning the inherent overhead of key negotiation and session establishment used in bidirectional protocols like quantum-resistant TLS [12] or SSH [13] is not tenable in C-V2X. More critically, the dynamic topology of vehicles on the road requires each vehicle to also periodically broadcast a certificate containing its public key so that constantly changing neighbors can verify its messages. The problem of frequently broadcasting huge QRC certificates is unique to CVs and, to date, has only been partially approached using a superseded set of QRC algorithms for now-outdated CV protocols that faced less crippling size constraints [14].

The restrictive cap on message size is partly due to the configuration of C-V2X at the lower layers, which even in the latest 5G standards is blind to external context and only makes decisions using some internal information. For example, under current standards, the modulation and coding scheme (MCS) that defines the data rate is conservatively adapted based solely on the amount of scheduled data and the speed of the sending vehicle [9]. The highly relevant signal-to-noise ratio (SNR) at nearby vehicles is disregarded, as it is considered too difficult to estimate in dynamic, broadcast-only

[1] Also known as post-quantum cryptography (PQC). We favor "quantum-resistant" as it more accurately reflects our intentions in this work.

Geoff Twardokus and Hanif Rahbari are with the ESL Global Cybersecurity Institute, Rochester Institute of Technology, Rochester, NY, USA (email: {geoff.twardokus,rahbari}@mail.rit.edu).

This work has been submitted to the IEEE for possible publication. Copyright may be transferred without notice, after which this version may no longer be accessible.

[2] 20 MHz bandwidth, with QPSK at a 0.3 code rate—see [8, Table 5.1.3.2-1].

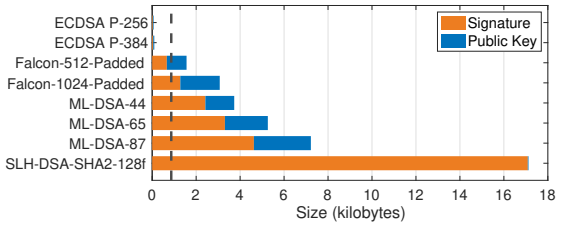


Fig. 1. Public key and signature sizes for ECDSA vs. NIST-approved QRC. Dashed line indicates payload constraint in a typical 20 MHz C-V2X channel.

V2V, and the focus is solely on reliability with no provision to balance it against potential *latency* that may undermine safety or security. As it is neither practical nor acceptable to redesign existing 4G and 5G C-V2X or to expect major changes in upcoming 6G standards, in this paper we advocate for a *more adaptive, context-aware* method of optimizing MCS selection that sits atop current 3GPP standards (and could integrate with future ones). Our technique exploits the awareness of surroundings provided by past safety messages to estimate the SNRs and more assertively prefer MCS options that effectively balance coverage and message size against message importance, allowing certain large messages to be sent faster to more relevant (e.g., closer) vehicles. That allows us to then develop an overarching fragmentation strategy that distributes large certificates quickly and securely to nearby vehicles, limiting message verification delays while de-prioritizing fast delivery to vehicles that represent a less immediate threat.

Our focus on 4G and 5G is vital because of the 13–15-year lifetime of an average vehicle [15], [16], the historical lack of consumer responsiveness to vehicle recalls [17], and the fact that 5G C-V2X equipment is only just reaching the commercial market [18]. These factors mean vehicles hitting the roads with 4G or 5G C-V2X equipment will be on the roads into the late 2030s, but will likely never be upgraded to 6G (which is not expected anyway until ~2032 [7]). Therefore, vehicles using 4G, 5G, and possibly 6G C-V2X will need to interoperate on the roads over the coming decade, and to support this 3GPP has gone as far as proposing new vehicles be equipped with both 4G and 5G modems to support long-term interoperability [19]. This also suggests that 6G C-V2X—though not yet finalized—is likely to follow a similar design to a refined 5G, in order to avoid the same kind of generational incompatibility within mixed fleets. A practical solution should bring QRC into C-V2X while also avoiding generational incompatibility—no small task, as this requires working within the restrictive bounds of 4G C-V2X that was never expected to handle the kind of overhead seen with QRC.

For similar reasons as those that require supporting multiple generations of C-V2X protocols, we must also move toward adopting QRC without immediately cutting off vehicles that can only use ECDSA. The safety-critical nature of C-V2X makes it imperative that no vehicles are prematurely excluded; however, this has to be balanced against the reality that classical signatures will eventually need to be removed from the system (e.g., NIST proposes disallowing ECDSA in 2035 [3]). An effective way to address this is with a *hybrid* protocol—one that supports both classical and quantum-resistant algorithms. The merits of hybrid approaches remain a topic of active debate, e.g., NIST does not always encourage them [20], whereas its German counterpart does [21]. Nevertheless, we argue that a hybrid approach is ideal for CVs. In the short term, legacy vehicles can continue to use ECDSA, while newer vehicles can use QRC instead of (or alongside) it. And in the long term, as legacy vehicles are upgraded or reach end of life (or ECDSA is disallowed), support for ECDSA can be smoothly removed by tweaking a protocol option to disallow ECDSA. The crypto-agile ability to switch without disruption is a critical feature for C-V2X security, as it allows supporting legacy vehicles for as long as needed without sacrificing quantum resistance of newer vehicles.

In this paper, we tackle the challenge of developing a *hybrid, quantum-resistant* protocol compatible with 4G, 5G, and (potentially) future 6G C-V2X that ❶ provides *provable* security guarantees against quantum-assisted spoofing and forgery attacks while ❷ retaining the classical security guarantees of current standards, ❸ further supports interoperability with legacy vehicles by not modifying the core C-V2X communication protocols, and ❹ does not compromise the speed or reliability of safety-critical C-V2X services. More specifically, we show a cross-layer adaptation can help, as:

- We augment existing C-V2X lower-layer functions in novel ways by developing intelligent, context-aware MCS selection based on a new technique for estimating SNR at distant receivers, containing the overhead of QRC without compromising strict latency and reliability requirements.
- We develop a novel security protocol called Certificate-based Hybrid Authentication of Safety Messages (`CHASM`) that brings quantum-resistance to C-V2X while carefully supporting cross-generational interoperability.
- We formally prove that integrating QRC into `CHASM` does not introduce new vulnerabilities and maintains strict security guarantees, confirming its robustness against message spoofing, forgery, and replay attacks.
- We experimentally demonstrate the practicality of `CHASM` for immediate deployment through both over-the-air hardware experiments and simulations, confirming that `CHASM` can support enough connected vehicles to handle expected CV density on the roads over the next 10–15 years without incurring significant performance penalties.

In the remainder of this paper, we first provide relevant background in Section II, followed by our system and threat models in Section III. Our `CHASM` protocol is presented in Section IV, enabled by our adaptive MCS scheme in Section V. In Section VI we formally prove the security of `CHASM` against classical and quantum adversaries. We provide experimental results in Section VII, discuss related work in Section VIII, and conclude with Section IX.

II. Technical Background

We now provide background on C-V2X, V2V security, and V2V safety that affects the design of our `CHASM` protocol.

A. Cellular Vehicle-to-Everything (C-V2X)

Standards for C-V2X are maintained by 3GPP in their Releases 14–15 for 4G, Releases 16–20 for 5G-Advanced,

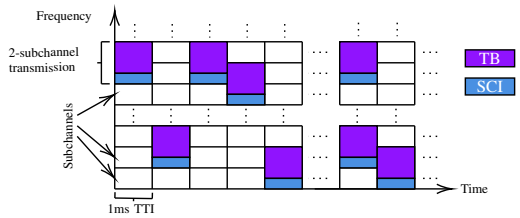


Fig. 2. PHY structure of an example C-V2X channel.

and planned Releases 21 and beyond for 6G [7]. C-V2X (in the context of this paper) is specifically focused on direct *sidelink* communication between vehicles (i.e., V2V), and in what follows we describe the design details of C-V2X sidelink that are common to V2V communication in both 4G and 5G C-V2X, which are extremely similar protocols.

1) Resource allocation: 4G and 5G C-V2X have two sidelink modes: "in-coverage", when a base station manages resource allocation, and "out-of-coverage," when a vehicle does not rely on cellular network service. The former helps (among other things) move the burden of medium sensing and transmission scheduling from vehicles to more resourceful base stations. However, the out-of-coverage mode is critical because it avoids making the availability of safety services dependent on cellular network service, whose quality may vary widely or be entirely unavailable (as of Dec. 2024, 32% of the U.S. still had no 4G coverage, and more than 50% had no 5G [22]). The key difference between these modes is whether a base station manages resource allocation or vehicles select resources autonomously, using a sensing protocol to avoid overlapping transmissions. We assume the more constrained out-of-coverage mode in this work, but note our protocol is fully compatible with the in-coverage mode.

2) Transmission size: Fig. 2 illustrates the physical (PHY) structure of the C-V2X channel for allocation. In time, the system defines 1 ms periods, each called a transmission time interval (TTI). Within each interval, the channel bandwidth is divided into resource blocks (RBs), which are grouped into equal-size *subchannels* for allocation. As an RB comprises an OFDMA structure with a fixed number of subcarriers and data symbols, the total number of RBs (and, thus, subchannels) is derived from channel bandwidth according to [23, Table 5.3.5-1]. We note that since RBs each have a fixed number of data symbols, for a given bandwidth and subcarrier spacing, MCS (see below) is the only additional factor in determining the maximum amount of data that can be transmitted in one TTI.

3) Transmitting sidelink data: To make a transmission, a vehicle first determines the MCS that it will use. As there is no bidirectional communication, traditional channel estimation (e.g., using uplink pilots in 5G) is inapplicable, and the MCS is instead selected from a few pre-configured options based on the transmitting vehicle's speed and the amount of data to be sent [9, Tables 11-12]. The vehicle then determines the number of RBs it will require, and accordingly determines the number of subchannels it needs (a minimum of two is required, and subchannels are never partially allocated [9]). Subsequently, the vehicle selects subchannels to use in an upcoming TTI, and it makes a transmission at a fixed power level (23 dBm) using these resources.

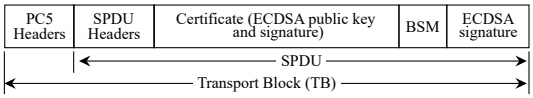


Fig. 3. Structure of an IEEE 1609.2 SPDU transmitted within a C-V2X transport block (note that PC5 headers are for C-V2X sidelink).

Each sidelink transmission is composed of two parts. First, two RBs are always used to transmit sidelink control information (SCI) that includes the metadata needed to recover the message payload. That payload is carried in an accompanying *transport block* (TB), as shown in Fig. 2. An important note for this paper is that the TB carries all of the application-layer data (e.g., a safety message and all security fields), which means the maximum size of a single C-V2X transmission is constrained by the maximum TB size of a given channel—e.g., in a 10 MHz channel, the maximum transmission size would be 48 RBs of data (from 50 total RBs minus 2 for SCI). Also note that the actual amount of data that can be carried depends on the selected MCS, *suggesting that a more carefully refined approach to MCS selection could yield a higher maximum transmission size without sacrificing reliability.* We explore this further in Section V.

B. Secure Exchange of Safety Messages

The V2V safety service is based on each vehicle periodically (every ~100 ms) broadcasting basic safety messages (BSMs) that convey its motion—location, speed, heading, etc.—to other vehicles within a ~600 m communication range. The mechanisms for verifying the authenticity and integrity of BSMs are defined by the IEEE 1609.2 standard [24], which provides for protecting BSMs by encapsulating them in larger secure protocol data units (SPDUs) that each contain a BSM, a digital signature over the BSM by the sending vehicle, and (sometimes) the public-key certificate needed to verify the sender's message signatures (see Fig. 3).

Certificates and their associated keys are managed through a public key infrastructure (PKI), such as the Security Credential Management System (SCMS) in the U.S. [25]. Such PKIs operate similarly to other systems—certificate authorities (CAs) issue certificates to end entities (e.g., vehicles), which can then use their certificates to sign messages. To preserve privacy, vehicles have several active certificates at a time, and they rotate the certificate used to sign messages every 5 minutes.

Receivers can verify a message's signature (and thus the authenticity and integrity of the message) by using the public key included in the sending vehicle's certificate. In order to facilitate this in a purely peer-to-peer network, each vehicle is required to periodically include its certificate in an outgoing SPDU, bundled with a periodic BSM. Under current standards [9], [26], only 20% of SPDUs (roughly every fifth transmission) carry the full certificate, while the remaining 80% instead carry a hash digest of the certificate. The hash can be used to look up the actual certificate (if the receiver already has it) and verify the associated message. This measure is intended to reduce the size of SPDUs and thereby free up bandwidth for others to use, while still ensuring minimal delay (<500 ms) for verifying BSMs that were signed using

that certificate. A key consideration in integrating QRC into C-V2X is to accommodate its overhead without adding significant message delivery or verification latency. We address this through our adaptive MCS selection.

C. MCS Selection for BSMs

When transmitting a BSM, a vehicle chooses the MCS to use based on just two factors: (1) the size (in bytes) of the message to be sent, and (2) the instantaneous speed of the vehicle [9, Tables 11–12]. This approach is used because in V2V, which is broadcast-only, there is no reciprocity or feedback[3] that can be used to estimate a channel between individual pairs of vehicles—i.e., the selection of MCS must be *channel- and receiver-agnostic*. Moreover, even though a vehicle can estimate the SNR at its own location, the lack of strong temporal or spatial correlation between rapidly moving vehicles means the SNR at one receiver has no relation to the SNR at other receivers, rendering local estimates unhelpful. As a result, the current approach prefers very conservative (i.e., low data-rate) MCS options in order to ensure higher reliability across all potential receivers.

The key limitation of this approach to MCS selection is that it does not take into consideration the varying importance of different messages to different nearby vehicles. For example, a message containing a warning that the sender's brakes have failed is much more important to vehicles that are nearby (perhaps within a few hundred meters) as opposed to vehicles that are a kilometer away—yet the current approach prefers to maximize the chances of *all* vehicles, not just the relevant ones, successfully receiving that message. We contend there is much room for improvement over this approach; in particular, we propose that broadening the range of factors considered in selecting an MCS—thus allowing messages to be treated differently based on their contents, rather than uniformly—will allow us to achieve a more practical and faster distribution of hybrid `CHASM` certificates, helping to mitigate the delay incurred by waiting to verify a BSM until a hybrid certificate is distributed in `CHASM`.

III. System and Threat Models

In the following, we present our system model, describe our threat model, and devise design requirements for our `CHASM` protocol.

A. System Model

The specific C-V2X technologies that we focus on, the operational assumptions we make about C-V2X environments, and the period of time we consider are as follows.

C-V2X Technology. Even though 5G C-V2X standards were published in 2020 (3GPP Rel. 16 [6]), devices on the market today continue to exclusively use 4G C-V2X for V2V services [18]. We anticipate greater uptake of 5G V2V in the near future. However, the 13-year average lifetime of a vehicle means 4G C-V2X equipment will be on the roads well into the 2030s, presumably alongside increasing numbers of 5G-capable vehicles. To support significant numbers of both 4G- and 5G-equipped vehicles over the next two decades, we design our protocol to be compatible with both protocols.

Operational Assumptions. Based on current spectrum allocations in the U.S. [27] and China [28], we assume a single 20 MHz channel is allocated for the C-V2X safety services. We also assume, based on current industry standards, that every vehicle is expected to broadcast a BSM every 100 ms [9], [26]. For security, we assume that all BSMs need to be verifiable using digital signatures backed by public-key certificates in an approach aligned with the most recent IEEE 1609.2 standard for V2V security [24]. We provide more specific security requirements below.

Operational Timeline. We focus on the next 10–20 years. This is a useful time period because it covers the evolution of C-V2X from 4G through 5G and into future 6G technology, as well as the time frame in which ECDSA will be disallowed and a quantum computer will likely be developed that can break today's cryptography [3]. We thus refer to the next 10–20 years as a *transition period* in which C-V2X needs to move, in a practical and non-disruptive manner, from today's completely classical cryptography to an eventual, completely quantum-resistant future. Fig. 4 illustrates these considerations and shows how `CHASM` adapts to their changes over time, which we will shortly elaborate on.

B. The Evolving Quantum Threat to C-V2X

The transition from classical to quantum-capable attackers will track the increasing capabilities of quantum computers, and we expect this to occur over a three-stage transition period (Fig. 4). In the classical stage, which runs up to today, a cryptographically relevant quantum computer (one that can easily break algorithms like ECDSA) does not exist, and the only plausible threats are from an adversary equipped with a classical computer. This is followed by the "Early Quantum Threat" stage in Fig. 4 that we are now entering, in which an attacker might have access to a modest, but likely not cryptographically relevant, quantum computer. This stage is marked by uncertainty. As quantum computing advances rapidly and hybrid classical-quantum approaches to breaking classical cryptography—known as "variational attacks" [29]—begin to show promise on even today's weak quantum computers [29], [30], one cannot remain certain that classical algorithms like ECDSA are still safe. Eventually, the third stage will be marked by emergence of a cryptographically relevant quantum computer that is known (or confidently assumed) to exist, making ECDSA unsafe and obsolete.

Given the uncertainty in exactly when a realistic quantum threat will emerge, we argue that the safety-critical nature of V2V means we cannot wait for the final stage (when a quantum threat is certain) and should instead begin moving to adopt QRC in C-V2X *today*. This is because, by design, the contents of BSMs—e.g., "a vehicle is about to run the red light in front of you"—will inform split-second decisions by human or autonomous drivers. Ensuring that BSMs are legitimate is therefore critical to ensure the driver's perception of their

[3]The physical feedback channel (PSFCH) introduced in 5G C-V2X applies only to unicast or multicast, which are not used for V2V safety services [19].

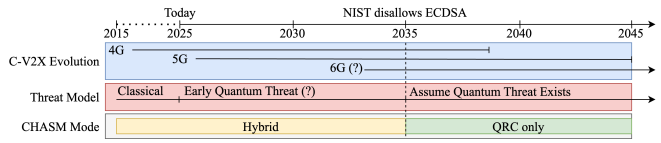


Fig. 4. CHASM is designed to keep up with expected evolution of C-V2X and increasing power of quantum computers over the next 10–15 years.

environment is accurate and any risky maneuvers (suddenly braking, swerving, etc.) are made only in response to actual threats. However, an adversary with access to a powerful quantum computer (or an effective variational approach) would significantly threaten that requirement.

For example, consider an attacker with access to a quantum computer that can break the 256-bit ECDSA keys and signatures used under IEEE 1609.2. Since other vehicles' public keys are broadcast alongside BSMs in unencrypted SPDUs (Fig. 3), such an attacker could harvest other vehicles' public keys and then use a quantum algorithm (e.g., Shor's [31]) to efficiently recover the corresponding private keys before they expire. Armed with those private keys, the attacker could then forge valid signatures on any bogus messages that they wish to inject into the environment, manipulating drivers' perceptions and tricking them into making unsafe decisions or possibly even crashing. Critically, since legitimate signatures and forgeries would be cryptographically indistinguishable, this attack would be extremely difficult to detect, severely compromising trust in C-V2X networks.

Considering these high stakes, we contend that just the *possibility* that an attacker might have access to such a quantum computer is sufficient to require upgrading to QRC—i.e., we cannot afford to wait until a concrete quantum threat becomes public before we take action. Hence, we design CHASM to move beyond ECDSA and introduce quantum-resistant security guarantees into C-V2X well before that day arrives, in the process overcoming the lack of preparedness to handle QRC overhead in current C-V2X systems (Fig. 1).

C. Threat Model

We consider an attacker who seeks to disrupt, delay, or otherwise degrade normal C-V2X operations in a target area, and we assume this attacker is able to send and receive standard C-V2X signals. We assume the attacker has access to computational capabilities consistent with the timeline described above and in Fig. 4—e.g., in the early quantum threat stage, attackers may or may not have access to quantum computing capabilities and/or variational techniques that degrade or break the effectiveness of ECDSA. We note that our design for CHASM, discussed next, accommodates this uncertainty. It can respond rapidly to unpredictable advances in the known or perceived quantum threat by easily toggling a protocol option to disable ECDSA—i.e., move to a fully quantum-resistant scheme—when, but only when, that is determined to be necessary.

D. Design Requirements for CHASM

In order to support both new and legacy equipment that will interoperate over the next several years, our protocol must:

1) Preserve existing (classical) security guarantees under IEEE 1609.2—namely, resistance to message spoofing and message forgery attacks.
2) Provide quantum-resistant security guarantees to match the classical guarantees of existing systems.
3) Support ECDSA for as long as possible while enabling a rapid and seamless switch to only using QRC once the decision to disallow ECDSA is taken.

Further, to avoid depending on a single QRC algorithm (which might harbor unknown vulnerabilities due to the relative immaturity of QRC), CHASM must:

4) Support multiple QRC algorithms and allow switching algorithms without major disruption; i.e., be *crypto-agile*.

Finally, in order to ensure that enhanced security does not compromise the safety functions of C-V2X, our protocol must:

5) Avoid degrading or interfering with the efficient and reliable operation of C-V2X safety services.

These requirements form the basis for the design of our CHASM protocol that is detailed in Section IV.

Security Requirements. We base our security requirements for CHASM on those enumerated in IEEE 1609.2—namely, that C-V2X messages must be protected against spoofing, alteration, and replay attacks [24, Sec. 1.2]. Protecting against these attacks requires authenticating messages, as well as verifying that they are received intact, and in CHASM we use digital signatures and message timestamps to accomplish this.

Privacy. Privacy concerns in V2V generally relate either to message contents or to vehicle (i.e., driver) location tracking. With respect to the former, we note that since BSM contents (position, direction of travel, speed, etc.) are not private and are not encrypted in commercial V2V systems, we do not consider message privacy (i.e., confidentiality) as a requirement for CHASM. As for location tracking, IEEE 1609.2 already provides a system of "pseudonymization" that cryptographically disconnects certificate public keys from permanent vehicle identities [24]. We note that CHASM is fully compatible with this system, which has been extended to QRC [32] and is deployed in current PKIs for V2V, and so we do not provide (redundant) privacy guarantees in the design of CHASM.

Scope. We limit our scope to the V2V exchange of safety messages between vehicles, as this is by itself a complicated problem to solve. We therefore make a few assumptions:

1) We consider CAs and the broader security infrastructure to be trusted and secure.
2) We assume all communication between CAs and vehicles occurs securely over separate communication channels (e.g., via a traditional cellular network) and cannot be observed or influenced by an adversary.
3) We assume the CA processes of generating and issuing certificates to vehicles are secure and cannot be affected by an adversary.
4) We exclude attacks that require physical access to vehicles or infrastructure (e.g., side-channels, tampering).

IV. CHASM: Certificate-based Hybrid Authentication of Safety Messages

For backwards compatibility and continuity with current standards during the transition to QRC, we design CHASM in

the spirit of current C-V2X security protocols: (1) by providing authentication by digitally signing all C-V2X messages, and (2) by supporting signature verification through digital certificates that are managed in a PKI. The critical difference is that CHASM adds quantum-resistant security to existing classical security guarantees, resulting in a flexible, *hybrid* protocol, and we propose modest adjustments to the design of IEEE 1609.2 certificates (Fig. 5) to support this.

A. Protocol Description

We now describe the specifics of how CHASM operates.

1) Certificate Fragmentation—Handling QRC Overhead: To protect BSMs against message spoofing or forgery attacks, every BSM needs to be sent in an SPDU that contains the BSM along with both a classical signature and a QRC signature over that BSM. Then, in order to verify those signatures, the sender of the BSM must supply receivers with the necessary classical and quantum-resistant public keys. It is straightforward to extend the digital certificates used today (Fig. 5(a)) to add the necessary quantum-resistant elements (Fig. 5(b)). However, when a hybrid certificate is constructed in this manner using any currently standardized QRC algorithm, the certificate cannot fit within the maximum size of a C-V2X transmission even by itself (see Fig. 1), let alone alongside the two message signatures and the actual BSM.

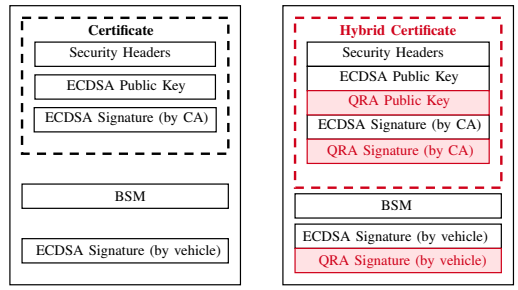
To get around this obstacle, we propose carefully *fragmenting* each certificate—i.e., rather than including the entire certificate in a single SPDU like in current systems (Fig. 3), we break the certificate into pieces and send each fragment in a separate SPDU, effectively spreading out the certificate across a few transmissions. Our intuition is that if one considers transmitting only the non-certificate elements shown in Fig. 5(b)—headers, a BSM, and two signatures—then in even the narrowest C-V2X channel, the time slot used for transmission will have some bandwidth "left over" that can be used to carry a certificate fragment. Note that the specific size of each fragment, and thus how many fragments are required to convey a complete certificate, will depend on both the choice of QRC (i.e., the actual size of the certificate) and the wireless parameters chosen for transmission. We will derive general expressions for this now, but will revisit how MCS parameters are selected in Section V.

If we denote a hybrid certificate by HC and an arbitrary fragment of that certificate by $\mathsf{HC_f}$, then the number of fragments required to transmit the complete certificate is simply the total certificate size divided by the maximum fragment size, or more formally by

$$n_\mathsf{f} = \left\lceil \frac{|\mathsf{HC}|}{\max|\mathsf{HC_f}|} \right\rceil \tag{1}$$

where n_f is the number of fragments and $|\cdot|$ is the "size of" operator, with the ceiling function $\lceil\cdot\rceil$ employed to ensure that n_f is a whole number. The numerator in (1) is trivially computed by summing the sizes of the individual hybrid certificate elements shown in Fig. 5(b), but deriving the maximum fragment size is slightly more involved.

In plain terms, the maximum fragment size is the difference between the maximum size of a C-V2X transmission and the total size of the non-certificate elements that are required for an SPDU (e.g., the BSM and signatures), as these determine how much bandwidth in a specific time slot will be "left over" for a fragment. Recalling Fig. 2 and 3, the maximum size of a transmission is equal to the largest TB that will encapsulate the transmitted SPDU; therefore, for a generic transport block (denoted by TB) and SPDU (denoted by S), we can express the maximum fragment size as:



(a) IEEE 1609.2-2022. (b) CHASM.

Fig. 5. Extension of classical SPDU security (a) to hybrid security (b) under CHASM. New elements to support quantum-resistant security are highlighted.

$$\max|\mathsf{HC_f}| = \max|\mathsf{TB}| - \min|\mathsf{S}|. \tag{2}$$

For a given C-V2X system, $\max|\mathsf{TB}|$ (in bytes) is determined by the modulation order (q), the code rate (r), and the number of RBs (N_{RB}) in the channel:

$$\max|\mathsf{TB}| = 12 \times 10 \times \log_2(q) \times r \times (N_{RB} - 2) \times \frac{1}{8} \tag{3}$$

The value of N_{RB} is a constant based on the channel bandwidth (B) as defined by [23, Table 5.3.5-1], while the values of q and r are determined by the transmitting vehicle[4] (we will elaborate shortly on how this is done). Then, the minimum size in bytes of a hybrid SPDU is

$$\min|\mathsf{S}| = |\mathsf{Hdr_{sec}}| + |\mathsf{M}| + \left|\mathsf{Sig}^c\right| + \left|\mathsf{Sig}^{qr}\right| \tag{4}$$

where $\mathsf{Hdr_{sec}}$ denotes IEEE 1609.2 security headers, M is a BSM, Sig^c is a classical signature, and Sig^{qr} is a quantum-resistant signature. Equations (1)–(4) together define the FRAGMENTCERTIFICATE procedure used to determine how many fragments are required for a particular hybrid certificate (HC), MCS (q, r), and bandwidth (B).

2) Transmitting Hybrid SPDUs: Algorithm 1 defines how SPDUs are constructed and transmitted while a specific end-entity certificate is in use (for roughly 5 minutes—see Section II). When certificate rotation is triggered, the algorithm simply restarts.

CHASM begins with loading pre-configured C-V2X parameters including the channel bandwidth, certificate transmission interval (denoted by T_c), and the required interval T_M between BSM transmissions (lines 1–3). The vehicle then loads its current certificate (line 4) and proceeds into the main execution loop (lines 5–22) which starts with setting runtime parameters and breaking the certificate (denoted by HC) into a suitable

[4]The constant values 12, 10, and 1/8 come from the OFDM structure of an RB in 5G C-V2X sidelink. Details can be found in [8, Table 5.1.3.2-1].

Algorithm 1 CHASM: Transmitting SPDUs

1: $B \leftarrow$ bandwidth from pre-configured C-V2X settings.
2: $T_{\mathsf{c}} \leftarrow$ certificate transmission interval (defaults to 500 ms).
3: $T_{\mathsf{M}} \leftarrow$ BSM interval (defaults to 100 ms).
4: $\mathsf{HC} \leftarrow$ current certificate
5: **while** $\mathsf{HC}^{\mathsf{id}}$ is active **do**
6: Select q and r values (the MCS to be used)
7: $\left(n_{\mathsf{f}}, \{\mathsf{HC}_1, \ldots, \mathsf{HC}_{n_{\mathsf{f}}}\}\right) \leftarrow$ FRAGMENT $\left(\mathsf{HC}^{\mathsf{id}}, q, r, B\right)$
8: $t_0 \leftarrow$ current time (in ms).
9: $i \leftarrow 1$
10: **for** $t \in \{t_0, t_0 + T_{\mathsf{M}}, \ldots, T_{\mathsf{c}}\}$ **do**
11: **if** $i \leq n_{\mathsf{f}}$ **then**
12: $\mathsf{Sig}^{c}, \mathsf{Sig}^{qr} \leftarrow$ signatures over $\mathsf{M} + \mathsf{HC}_i$ using $\mathsf{sk}^{c}_{\mathsf{id}}, \mathsf{sk}^{qr}_{\mathsf{id}}$
13: $\mathsf{S}_i \leftarrow$ SPDU containing $\mathsf{M}, \mathsf{HC}_i, \mathsf{Sig}^{c}, \mathsf{Sig}^{qr}$
14: Transmit S_i
15: $i = i + 1$
16: **else**
17: $\mathsf{Sig}^{c}, \mathsf{Sig}^{qr} \leftarrow$ signatures over M using $\mathsf{sk}^{c}_{\mathsf{id}}, \mathsf{sk}^{qr}_{\mathsf{id}}$
18: $\mathsf{S} \leftarrow$ SPDU containing $\mathsf{M}, \mathsf{Sig}^{c}, \mathsf{Sig}^{qr}$
19: Transmit S
20: **end if**
21: **end for**
22: **end while**

Algorithm 2 CHASM: Receiving and Verifying SPDUs

1: **procedure** PROCESSSPDU(S^*)
2: $\mathsf{id} \leftarrow$ extract vehicle ID from S^*
3: **if** id corresponds to a known, verified certificate **then**
4: $\mathsf{HC}^{\mathsf{id}} \leftarrow$ load known certificate
5: VERIFYSPDU $\left(\mathsf{HC}^{\mathsf{id}}, \mathsf{S}\right)$
6: **else**
7: **if** S contains a certificate fragment $\mathsf{HC}^{\mathsf{id}}_{\mathsf{f}}$ **then**
8: Cache $\mathsf{HC}^{\mathsf{id}}_{\mathsf{f}}$ for reassembly after all fragments are received.
9: **end if**
10: Cache S for processing once certificate is obtained and verified.
11: **if** all fragments for certificate of id have been received **then**
12: $\mathsf{HC}^{\mathsf{id}} \leftarrow$ reassemble certificate fragments for id
13: $\mathsf{r} \leftarrow$ VERIFYCERTIFICATE $\left(\mathsf{HC}^{\mathsf{id}}\right)$
14: **if** r == True **then**
15: Record $\mathsf{HC}^{\mathsf{id}}$ as known and valid certificate for id
16: **for** each cached SPDU S^* received from id **do**
17: VERIFYSPDU ($\mathsf{id}, \mathsf{S}^*$)
18: **end for**
19: **end if**
20: **end if**
21: **end if**
22: **end procedure**

23: **procedure** VERIFYSPDU($\mathsf{HC}^{\mathsf{id}}, \mathsf{S}^*$)
24: $\left(\mathsf{pk}^{c}_{\mathsf{id}}, \mathsf{pk}^{qr}_{\mathsf{id}}\right) \leftarrow$ extract public keys from $\mathsf{HC}^{\mathsf{id}}$
25: Verify signatures of S^* using $\left(\mathsf{pk}^{c}_{\mathsf{id}}, \mathsf{pk}^{qr}_{\mathsf{id}}\right)$
26: **if** verification of signatures for S^* is successful **then**
27: **Accept** S
28: **else**
29: **Reject** S^*
30: **end if**
31: **end procedure**

32: **procedure** VERIFYCERTIFICATE($\mathsf{HC}^{\mathsf{id}}$)
33: Verify certificate using PKI and trust root as per IEEE 1609.2 [24].
34: **if** verification is successful **then**
35: **return** True
36: **else**
37: **return** False
38: **end if**
39: **end procedure**

number of fragments (lines 6–7). After fragmenting the certificate, a timer is initialized to the start time (t_0), and a counter variable (i) is defined to track fragment transmissions, after which the main execution loop is reached.

The loop defined from lines 10–21 controls the transmission of SPDUs as well as the specific contents of each transmission. The loop repeats according to the configured BSM interval—i.e., if $T_{\mathsf{M}} = 100$ ms, then the loop executes every 100 ms to ensure BSMs are sent at the appropriate rate. For each transmission (i.e., each iteration of the loop), an SPDU is constructed and transmitted, with the first n_{f} SPDUs containing sequential certificate fragments. Once all fragments are transmitted, transmission continues every T_{M} ms until the next certificate transmission time ($t_0 + T_{\mathsf{c}}$) with SPDUs that each contain a BSM, a hash of the certificate (see Section II), and the associated signatures.

3) Receiving SPDUs: When an unverified SPDU (denoted by S^*) is received, a vehicle executes the PROCESSSPDU procedure described in Algorithm 2. First, the sender (denoted by id) is identified (line 2), either using the certificate fragment contained in the SPDU (if one is present) or using the included hash digest of the sender's certificate (see Section II-B). If the certificate for id is already known (i.e., it was previously obtained and has been verified as described below), then the SPDU can be verified (or rejected) by calling the VERIFYSPDU procedure described shortly. Otherwise, if a certificate fragment is contained in S, then it is stored for later reassembly (line 8) and the SPDU is cached (line 10) so that its encapsulated BSM can be verified later, once the complete certificate has been obtained and verified. At this point, a check is performed to see if all of the fragments for id's certificate have been received (line 11), and if so, the certificate is reassembled and verified by calling the VERIFYCERTIFICATE procedure described below. If the certificate is successfully verified, then the certificate is recorded for id as known and verified (line 15), after which any cached SPDUs signed using that certificate can be processed (lines 16–18).

a) VERIFYSPDU*:* Given an unverified SPDU (denoted S^*) and an authenticated certificate (denoted $\mathsf{HC}^{\mathsf{id}}$), the BSM contained in the SPDU is verified by extracting the classical and quantum-resistant public keys from the certificate (line 24), and then using them to verify the corresponding signatures over the BSM (line 25). Finally, the results of signature verification determine whether or not to accept the BSM as authentic (lines 26–30).

b) VERIFYCERTIFICATE*:* A hybrid certificate for claimant id ($\mathsf{HC}^{\mathsf{id}}$) is verified along the lines of a typical PKI. First, the classical and quantum-resistant signatures on $\mathsf{HC}^{\mathsf{id}}$ by the issuing CA are verified using the CA's public keys, which are obtained from the certificate issued to that CA by a higher-tier CA, and so on following the *chain of trust* until a trusted root authority is reached. For brevity, we do not go into detail on this process here, and we assume that (1) the vehicle has (or can easily obtain) any CA's certificate, and (2) the CA certificate is authenticated.

V. Context-Aware Adaptive Modulation for Accelerated Certificate Distribution

Fragmentation inherently increases the amount of time that is required to complete the transmission of a given certificate. Until a vehicle's certificate is fully assembled, receiving vehicles cannot verify its BSMs (line 11 of Algorithm 2). The larger the number of fragments as a result of larger keys or signatures, especially in busier channels where bandwidth is in high demand (especially on dense roadways), the longer the resulting delays in BSM verification will be, possibly causing the cached BSMs to expire before they can be processed. Clearly, we need to resolve this problem, or else CHASM would adopt QRC at the cost of potentially crippling the V2V safety service. In this section, we develop a solution based on enhancing MCS selection in V2V, leveraging the inherent situational awareness of connected vehicles to facilitate a more intelligent and adaptive approach.

A. Higher MCS for Certificates as a Pragmatic Trade-off

The preference for lower MCS values in current C-V2X (see Section II) is predicated on the extreme assumption that all messages are equally important to all vehicles in communication range. This dates from early V2V standards and aligned with expectations of that time (the mid 2010s), but we now question it—especially from a security perspective.

Recalling that the primary safety case for V2V is avoiding vehicle collisions, combined with the reality that vehicles which are closer together are at a greater risk of colliding with each other, then it follows that *vehicles in closer proximity to each other can tolerate less delay in verifying and processing BSMs.* In turn, this means that it is more important for nearby vehicles to receive a certificate quickly (minimizing delay to BSM verification) than for all vehicles in communication range to reliably receive the certificate. Transmitting SPDUs that contain certificate fragments at a higher MCS would allow for larger fragments to be sent, conveying the complete certificate to nearby vehicles in fewer transmissions, while accepting that vehicles which are further away may not be able to receive the certificate fragments. Also, using higher MCS options to transmit certificates enables CHASM to support a wider range of QRC algorithms, enhancing its crypto-agility. As Table I shows, higher MCS values allow more algorithms to satisfy the current standards' requirement of delivering a complete certificate in under 500 ms.

TABLE I
Certificate fragment count to convey a full certificate using different QRC under varying MCS in a 20 MHz channel.

	QPSK 0.30	QPSK 0.50	QPSK 0.66	16-QAM 0.36
Falcon-512-Padded	21	3	2	2
ML-DSA-44	48	6	4	4
ML-DSA-65	67	8	5	6

B. Context-Aware Adaptive Modulation

An effective MCS selection for a given transmission should more aggressively target higher data rates when certain important messages (e.g., those containing certificate fragments) need to be conveyed more quickly to the closer vehicles. Therefore, we develop an approach that allows a transmitter to estimate a critical distance, based on message contents, within which a certain outage probability is satisfied. Utilizing its knowledge of nearby vehicles' locations (obtained from their BSMs at the upper layers), the transmitter estimates average SNR across nearby vehicles—which cannot be directly calculated in C-V2X (see Section II)—and then calculates a link budget margin based on the desired outage probability. Finally, the transmitter selects the most suitable MCS under those constraints. This process is detailed in the following.

For N potential receivers within the critical distance d, the mean SNR at those receivers *without fading* can be estimated for a receiver $i \in \{1, \ldots, N\}$ as:

$$\bar{\gamma}_i = P_t + G_t + G_r - PL\left(d_i\right) - N_0\left(B\right) \tag{5}$$

where $\bar{\gamma}$ is the estimated mean SNR, P_t is the transmit power in dBm, while G_t and G_r are the transmitter and receiver antenna gains in dBi. $PL\left(d_i\right)$ is the log-normal path loss in terms of the path-loss exponent α and path loss PL_0 at reference distance of 1 m as

$$PL(d) = PL_0 + 10\alpha \log_{10}\left(d\right) \tag{6}$$

while $N_0\left(B\right)$ is the noise power in a channel of bandwidth B.

With (5), the outage probability is equally distributed between independent probabilities of loss due to fading or shadowing, which will be used to determine the necessary fading margin (denoted by $M_f\left(\epsilon_f\right)$) and shadowing margin (denoted by $M_s\left(\epsilon_s\right)$). With ϵ denoting the total acceptable outage probability:

$$\epsilon = \epsilon_f + \epsilon_s \text{ where } \quad \epsilon_f = \epsilon_s \tag{7}$$

For fast-fading in a Rayleigh channel, the outage probability is well-known [33] as the expression

$$P_{out,fast} = 1 - \exp\left(\frac{-\beta_m}{\bar{\gamma}}\right) \tag{8}$$

where β_m is the required SNR to meet the targeted block error rate for a candidate MCS denoted by m. By then setting $P_{out,fast} = \epsilon_f$ and denoting the fading margin by

$$M_f\left(\epsilon_f\right) = \frac{\bar{\gamma}}{\beta_m}$$

to indicate the increase in SNR needed to compensate for Rayleigh fading dips, we can solve for the required fading margin in dB:

$$M_f\left(\epsilon_f\right) = 10\log_{10}\left(\frac{1}{-\ln\left(1-\epsilon_f\right)}\right). \tag{9}$$

For shadowing, we assume a Gaussian distribution with variance σ_s: $X_s \sim \mathcal{N}\left(0, \sigma_s^2\right)$. We require the outage probability to fall below the targeted shadowing loss probability ϵ_s:

$$P\left(X_s < -M_s\right) = \epsilon_s. \tag{10}$$

This allows us to express the shadowing margin M_s as:

$$M_s\left(\epsilon_s\right) = \sigma_s * Q^{-1}\left(1-\epsilon_s\right). \tag{11}$$

At last, the transmitter can use the estimated receiver SNRs from (5) in combination with the fading margins given by (9) and (11) to determine the probability that each receiver i within the critical distance d will successfully recover the transmission. We assume receiver i can recover the transmission if the mean SNR at i meets or exceeds the sum of the required SNR for MCS m and the margins:

$$\mu_{\gamma,i} \geq \beta_m + M_f\left(\epsilon_f\right) + M_s\left(\epsilon_s\right). \tag{12}$$

Then, for a candidate MCS m, the proportion of receivers within the critical distance that are predicted to successfully decode the message is given by

$$p\left(m\right) = \frac{1}{N_d}\sum_{i=1}^{N}\mathcal{I}\left\{\mu_{\gamma,i} \geq \beta_m + M_f\left(\epsilon_f\right) + M_s\left(\epsilon_s\right)\right\} \tag{13}$$

where N_d is the total number of receivers within d meters of the transmitter and $\mathcal{I}\{\cdot\}$ is an indicator function that returns 1 if $\cdot$, else 0. After evaluating (13) for all candidate MCSs, the transmitter chooses the highest candidate MCS for which

$$p\left(m\right) \geq p_{target} \tag{14}$$

is true, where p_{target} is the proportion of receivers within distance d required to be successful recipients.

VI. Formal Security Analysis

We now formally prove the security of CHASM against message spoofing and message forgery attacks by showing that CHASM is EUF-CMA secure under the assumption that the algorithms used to instantiate the protocol have that property.

A. Notations and Definitions

In our proofs, signatures are denoted by Sig, public keys by pk, and secret (private) keys by sk. In general, we use subscripts to indicate specific properties and superscripts to distinguish classical and quantum-resistant elements. For example, a classical public key belonging to a vehicle with the identifier XYZ would be denoted as $\mathsf{pk}^c_{\mathsf{XYZ}}$, while a quantum-resistant digital signature by a certificate authority would be denoted by $\mathsf{Sig}^{qr}_{\mathsf{CA}}$.

Definition 1 (Digital Signature Scheme)**.** A digital signature scheme Σ is a tuple of polynomial-time algorithms $\Sigma = (\mathsf{KG}, \mathsf{SG}, \mathsf{V})$ where:

- $\{\mathsf{sk}, \mathsf{pk}\} \leftarrow \mathsf{KG}\left(1^\lambda\right)$: **key generation** takes security parameter λ, returns pair of secret and public keys $\{\mathsf{sk}, \mathsf{pk}\}$.
- $\mathsf{Sig} \leftarrow \mathsf{SG}\left(\mathsf{sk}, m\right)$: **signature generation** takes secret key sk and message $m \in \mathcal{M}$, and returns signature Sig.
- $\mathsf{r} \leftarrow \mathsf{V}\left(\mathsf{pk}, m, \mathsf{Sig}\right)$: **signature verification** takes public key pk, message m, signature Sig, and returns $\mathsf{r} \in \{0, 1\}$, where 1 is verification success and 0 is failure.

This definition is typical for digital signature schemes.

Definition 2 (Dual EUF-CMA Unforgeability Experiment)**.** We define the *dual unforgeability experiment*, denoted by $\mathsf{Expt}^{\mathsf{eUF}}_{\mathsf{dual}}$, in which the challenger generates two keypairs (one classical and one quantum-resistant), and an adversary $\mathcal{A}$ is then provided with the public (but not secret) keys, denoted pk^c and pk^{qr}. We assume adversary $\mathcal{A}$ has access to query two signing oracles $\mathcal{O}^c_{\mathsf{sign}}\left(\mathsf{m}, \mathsf{sk}^c\right)$ and $\mathcal{O}^{qr}_{\mathsf{sign}}\left(\mathsf{m}, \mathsf{sk}^{qr}\right)$ with an arbitrary message m. The oracles operate in a black-box manner and respectively return signatures Sig^c and Sig^{qr} over m using the appropriate secret key (which $\mathcal{A}$ does not have access to). $\mathcal{A}$ wins if it outputs a tuple $\left(\mathsf{m}^\star, \mathsf{Sig}^{c\star}, \mathsf{Sig}^{qr\star}\right)$ such that either:

1) $\mathsf{V}^c\left(\mathsf{pk}^c, \mathsf{m}^\star, \mathsf{Sig}^{c\star}\right) = 1$; or
2) $\mathsf{V}^{qr}\left(\mathsf{pk}^{qr}, \mathsf{m}^\star, \mathsf{Sig}^{qr\star}\right) = 1$.

Note that this requires *at least one* signature to be valid; i.e., the adversary wins if *either* signature can be forged.

Assumption: EUF-CMA for Signature Algorithms. Our reduction is a black-box approach with respect to the digital signature algorithms that are actually used. We rely on the assumption that standard proofs can be found elsewhere for the EUF-CMA security of the underlying algorithms, and we provide security reductions to those properties in order to prove the essential requirement that the particular operations of CHASM do not introduce new vulnerabilities when using otherwise secure algorithms.

Assumption: PKI. In our proofs, we rely on well-established principles of certificate-based authentication in a PKI. In particular, we assume that there is a root CA which is implicitly trusted and whose public key $\mathsf{pk}_{\mathsf{root}}$ is known to all vehicles. We assume this public key cannot be modified or tampered with by any adversary, and we assume that the corresponding secret key $\mathsf{sk}_{\mathsf{root}}$ (or keys, in a hybrid design) cannot be obtained by the adversary. These assumptions facilitate constructing *authenticated certificate chains*; i.e., the root CA signs the certificates of its direct subordinate CAs, those CAs sign their subordinates' certificates, etc., until a CA issues a certificate to a vehicle. On this basis, we assume in our proofs that when a vehicle is attempting to authenticate the certificate belonging to a particular vehicle, it is able to authenticate the issuing CA's certificate using the associated certificate chain back to the trusted root CA. As described in Section III, attacks targeting CAs are out of scope.

B. Security Proofs

We now introduce our core theorems and corollaries. We prove their security against both classical and quantum-capable adversaries, thus proving CHASM is secure across all stages of the transition to QRC (Fig. 4).

1) Hybrid Certificate Security Under Fragmentation: We first prove hybrid certificates in CHASM are unforgeable even when fragmented. We assume the adversary here has full control over the network and can drop, reorder, replay, or inject fake fragments. We further assume that the attacker does not have access to the CA's secret keys, but they have access to classical and quantum-resistant signing oracles that produce valid CA signatures over arbitrary queries.

Definition 3 (Authenticity Under Fragmentation)**.** We define the *authenticity under fragmentation* experiment, denoted by $\mathsf{ExpAUF}\left(\mathcal{A}\right)$, as follows:

1) The challenger generates CA keypairs ($\mathsf{pk}^c_{\mathsf{CA}}$,$\mathsf{sk}^c_{\mathsf{CA}}$) and ($\mathsf{pk}^{qr}_{\mathsf{CA}}$,$\mathsf{sk}^{qr}_{\mathsf{CA}}$), and then supplies the public keys to $\mathcal{A}$.

2) $\mathcal{A}$ outputs a set of hybrid certificate fragments and supplies them to the verifier.
3) The verifier reassembles the fragments in the indicated order and verifies both the classical and quantum-resistant signatures.
4) $\mathcal{A}$ wins ($\mathsf{ExpAUF} = 1$) if the verifier accepts a certificate that was not signed by either signing oracle.

Theorem 1 (AUF Security)**.** Given a probabilistic polynomial-time adversary $\mathcal{A}$ with access to classical and/or quantum computing power, if both the classical and quantum-resistant signature schemes used in CHASM are EUF-CMA secure, then

$$\Pr\left[\mathsf{ExpAUF}\left(\mathcal{A}\right) = 1\right] \leq 2{*}\max\{Adv^{\mathsf{C}}_{\mathsf{eUF}}\left(\mathcal{B}_1\right), Adv^{\mathsf{Q}}_{\mathsf{eUF}}\left(\mathcal{B}_2\right)\}$$

where $\mathcal{B}_1, \mathcal{B}_2$ are (respectively) classical and quantum-enabled EUF-CMA forgers with similar efficiency to $\mathcal{A}$.

Proof: Assume for a contradiction that there exists a probabilistic polynomial-time adversary $\mathcal{A}$ who can win $Pr\left[\mathsf{ExpAUF}\left(\mathcal{A}\right) = 1\right]$ with non-negligible probability. We begin by flipping a fair coin $b \in \{0, 1\}$ to choose the algorithm $\mathcal{A}$ will attempt to forge a signature for, and we proceed along one of two branches based on the result.

Case 0 (Classical Forger): We transform $\mathcal{A}$ into the classical forger $\mathcal{B}_1$. The classical public key $\mathsf{pk}^{c*}_{\mathsf{id}}$ is obtained from the challenger and we embed this as the CA's public key. The adversary can query the classical signing oracle, while quantum-resistant queries are answered honestly by the simulator (who knows the quantum-resistant secret key $\mathsf{sk}^{qr}_{\mathsf{CA}}$). Eventually, $\mathcal{B}_1$ outputs a classical signature that does not appear in the signing oracle's transcript (i.e., a previously unseen signature). If that signature verifies with $\mathsf{pk}^{c*}_{\mathsf{id}}$, $\mathcal{B}_1$ wins.

Case 1 (Quantum Forger): This case is the parallel construction to Case 0 using quantum forger $\mathcal{B}_2$. We omit a complete description as it is redundant with Case 0, but observe that this case yields the parallel result that $\mathcal{B}_2$ wins by producing a valid, previously unseen quantum-resistant signature.

Conclusion: If $\mathcal{A}$ makes the verifier accept a forged certificate, then by construction both of the signatures on that certificate must be verifiable with the issuing CA's public keys. Moreover, at least one signature must be fresh (i.e., unseen by any oracle) because only one oracle is ever active at a time. In Case 0, the fresh signature is classical, which means adversary $\mathcal{B}_1$ has broken EUF-CMA for the classical algorithm—a contradiction. In Case 1, the fresh signature is quantum-resistant, which means adversary $\mathcal{B}_2$ has broken EUF-CMA for the quantum-resistant algorithm—also a contradiction. As both cases arrive at a contradiction, we conclude CHASM is AUF-secure if its underlying algorithms are EUF-CMA secure □.

2) Hybrid SPDU Unforgeability:

Theorem 2 (EUF-CMA Security—Classical Adversary)**.** Given an adversary $\mathcal{A}^c$ who only has access to classical computing powers, CHASM is EUF-CMA secure if the classical and quantum-resistant algorithms used to instantiate the protocol are both EUF-CMA secure. More formally:

$$Adv^{\mathsf{eUF}}_{\mathsf{CHASM}}\left(\mathcal{A}^C\right) \leq 2 * \max\left(Adv^{\mathsf{C}}_{\mathsf{eUF}}\left(\mathcal{A}^C\right), Adv^{\mathsf{Q}}_{\mathsf{eUF}}\left(\mathcal{A}^C\right)\right)$$

where $Adv^{\mathsf{eUF}}_{\mathsf{CHASM}}\left(\mathcal{A}^C\right)$ is the advantage of classical adversary $\mathcal{A}^C$ against CHASM, $Adv^{\mathsf{C}}_{\mathsf{eUF}}\left(\mathcal{A}^C\right)$ is the advantage of $\mathcal{A}^C$ against the classical algorithm and $Adv^{\mathsf{Q}}_{\mathsf{eUF}}\left(\mathcal{A}^C\right)$ is the advantage of $\mathcal{A}^C$ against the quantum-resistant algorithm.

Proof: Let S be an SPDU that contains a BSM (denoted by M), the sender's public keys $\mathsf{pk}^c_{\mathsf{id}}$ and $\mathsf{pk}^{qr}_{\mathsf{id}}$, and the signatures $\mathsf{Sig}^c_{\mathsf{id}}$ and $\mathsf{Sig}^{qr}_{\mathsf{id}}$ over M. The SPDU

$$\mathsf{S} = \left(\mathsf{M}, \mathsf{pk}^c_{\mathsf{id}}, \mathsf{pk}^c_{\mathsf{id}}, \mathsf{pk}^{qr}_{\mathsf{id}}, \mathsf{Sig}^{qr}_{\mathsf{id}}\right)$$

is then sent to a receiver. The individual signatures are verified at the receiver using the verification algorithms given in Def. 1:

$$\mathsf{r}^c_{\mathsf{bsm}} \leftarrow \mathsf{BSMVrfy}\left(\mathsf{pk}^c_{\mathsf{id}}, \mathsf{Sig}^c_{\mathsf{id}}, \mathsf{M}\right)$$

and

$$\mathsf{r}^{qr}_{\mathsf{bsm}} \leftarrow \mathsf{BSMVrfy}\left(\mathsf{pk}^{qr}_{\mathsf{id}}, \mathsf{Sig}^{qr}_{\mathsf{id}}, \mathsf{M}\right)$$

where $\mathsf{r}^c_{\mathsf{spdu}}, \mathsf{r}^{qr}_{\mathsf{spdu}} \in \{0, 1\}$ and 1 indicates verification success, and 0 indicates failure. The final verification result is

$$\mathsf{r}_{\mathsf{final}} \leftarrow \mathsf{r}^c_{\mathsf{bsm}} \vee \mathsf{r}^{qr}_{\mathsf{bsm}} \tag{15}$$

where $\mathsf{r}_{\mathsf{final}} = 1$ is success, otherwise failure (note (15) is an OR because $\mathsf{r}^{qr}_{\mathsf{bsm}}$ can return $\perp$ on legacy vehicles that do not support QRC). This proof is intended for the period of time before classical algorithms are disallowed (i.e., there is no known quantum threat), and thus $\mathsf{r}^c_{\mathsf{bsm}} = 1$ can only occur if the classical algorithm is compromised by a classical adversary, which violates EUF-CMA for the classical algorithm.

Our proof proceeds by assuming for a contradiction that there is an efficient classical adversary $\mathcal{A}^c$ who can win the $\mathsf{Expt}^{\mathsf{eUF}}_{\mathsf{dual}}$ with non-negligible probability ϵ. We then flip a fair coin to determine which algorithm the adversary attacks and branch accordingly. Queries by the adversary are forwarded only for the selected algorithm, and other queries return $\perp$, so that the adversary winning in the selected branch implies a win probability of at least $1/2$. Each branch is as follows:

Case 0: The challenger provides $\mathcal{A}^c$ with the sender's classical public key $\mathsf{pk}^c_{\mathsf{id}}$ (e.g., from the verified certificate associated with M). $\mathcal{A}^c$ can query the classical signing oracle instantiated as $O^c_{\mathsf{sign}}\left(\cdot, \mathsf{sk}^c_{\mathsf{id}}\right)$, and the result of any query is stored in a list of tuples $(\mathsf{m}^\star, \mathsf{Sig}^c_{\mathsf{id}})$ where $\mathsf{m}^\star$ is the queried message and $\mathsf{Sig}^c_{\mathsf{id}}$ is the signature over $\mathsf{m}^\star$ calculated using the real secret key $\mathsf{sk}^c_{\mathsf{id}}$ within the oracle (although, critically, recall that $\mathcal{A}^c$ does not have access to that secret key). The classical signing oracle is simulated on input $\mathsf{m}^\star$ as:

1) $\mathsf{Sig}^{c\star}_{\mathsf{id}} \leftarrow \mathsf{SG}^c\left(\mathsf{sk}^c_{\mathsf{id}}, \mathsf{m}^\star\right)$
2) Return $\left(\mathsf{Sig}^{c\star}\right)$ to $\mathcal{A}^c$

Eventually, $\mathcal{A}^c$ produces output $(\mathsf{m}^\star, \mathsf{Sig}^{c\star}_{\mathsf{id}})$, where $\mathsf{Sig}^{c\star}_{\mathsf{id}}$ comes from the signing oracle, such that $\mathsf{Sig}^{c\star}_{\mathsf{id}}$ is verifiable with $\mathsf{pk}^c_{\mathsf{id}}$. With this output, either $\mathsf{m}^\star$ was never queried, in which case $\mathsf{Sig}^{c\star}_{\mathsf{id}}$ is a valid forgery for a new message, or $\mathsf{m}^\star$ was queried previously but the new signature $\mathsf{Sig}^{c\star}_{\mathsf{id}}$ does not match the stored signature from the previous query, in which case $\mathsf{Sig}^{c\star}_{\mathsf{id}}$ is a valid (new) forged signature for a pre-existing message. In either case, $\mathcal{A}^c$ breaks EUF-CMA for the classical signature algorithm, a contradiction.

Case 1: The challenger provides $\mathcal{A}^c$ with the sender's quantum-resistant public key (again, from a verified certificate). $\mathcal{A}^c$ can query a quantum-resistant signing oracle instantiated as $O^{qr}_{\mathsf{sign}}(\cdot, \mathsf{sk}^{qr}_{\mathsf{id}})$, and the result of any query is stored in a list of tuples $(\mathsf{m}^\star, \mathsf{Sig}^{qr}_{\mathsf{id}})$ where $\mathsf{m}^\star$ is the queried message and $\mathsf{Sig}^{qr}_{\mathsf{id}}$ is the signature over $\mathsf{m}^\star$ calculated using the real secret key $\mathsf{sk}^{qr}_{\mathsf{id}}$ within the oracle. The quantum-resistant signing oracle is simulated on input $\mathsf{m}^\star$ as:

1) $\mathsf{Sig}^{qr\star}_{\mathsf{id}} \leftarrow \mathsf{SG}^{qr}\left(\mathsf{sk}^{qr}_{\mathsf{id}}, \mathsf{m}^\star\right)$
2) Return $\left(\mathsf{Sig}^{qr\star}\right)$ to $\mathcal{A}^c$

Eventually, $\mathcal{A}^c$ produces output $(\mathsf{m}^\star, \mathsf{Sig}^{qr\star}_{\mathsf{id}})$, where $\mathsf{Sig}^{qr\star}_{\mathsf{id}}$ is obtained from the signing oracle, such that $\mathsf{Sig}^{qr\star}_{\mathsf{id}}$ is verifiable with $\mathsf{pk}^{qr\star}_{\mathsf{id}}$. With this output, either $\mathsf{m}^\star$ was never queried, in which case $\mathsf{Sig}^{qr\star}_{\mathsf{id}}$ is a valid forgery for a new message, or $\mathsf{m}^\star$ was queried previously but the new signature $\mathsf{Sig}^{qr\star}_{\mathsf{id}}$ does not match the stored signature from the previous query, in which case $\mathsf{Sig}^{qr\star}_{\mathsf{id}}$ is a valid (new) forged signature for a pre-existing message. In either case, $\mathcal{A}^c$ breaks the EUF-CMA security of the quantum-resistant signature algorithm, a contradiction.

Conclusion: If $\mathcal{A}^c$ can generate either $(\mathsf{m}^\star, \mathsf{Sig}^{c\star}_{\mathsf{id}})$ or $(\mathsf{m}^\star, \mathsf{Sig}^{qr\star}_{\mathsf{id}})$ where the respective, legitimate public keys $\mathsf{pk}^c_{\mathsf{id}}$ or $\mathsf{pk}^{qr}_{\mathsf{id}}$ successfully verify, then $\mathcal{A}^c$ forged either a classical or quantum-resistant signature over M. This means $\mathcal{A}^c$ broke EUF-CMA for one of the algorithms used in CHASM, a contradiction under our assumption that both algorithms are EUF-CMA secure. Therefore, the actual advantage of $\mathcal{A}^c$ against each algorithm is negligible, which means the advantage against CHASM must also be negligible, and thus CHASM is EUF-CMA secure against classical adversaries. □

Corollary 2.1 (EUF-CMA Security—Quantum Adversary)**.** Once it is known (or assumed) that ECDSA can be broken by a quantum-equipped adversary, CHASM will drop support for authentication using classical algorithms. Formally, this means we modify Theorem 2 by replacing classical adversary $\mathcal{A}^C$ with quantum-capable adversary $\mathcal{A}^Q$, and further by requiring only unforgeability in the quantum-resistant branch (Case 1). We accordingly derive that given an adversary $\mathcal{A}^Q$ who has access to both quantum and classical computing powers, CHASM is EUF-CMA secure if the quantum-resistant algorithm used to instantiate the protocol is EUF-CMA secure:

$$Adv^{\mathsf{eUF}}_{\mathsf{CHASM}}\left(\mathcal{A}^Q\right) \leq Adv^{\mathsf{Q}}_{\mathsf{eUF}}\left(\mathcal{A}^Q\right).$$

Critically, we note this removes any dependence on classical algorithms from the EUF-CMA proof for CHASM, thus ensuring, in plain terms, that as long as the quantum-resistant algorithm is secure, CHASM is secure against a quantum adversary, irrespective of the use or validity of ECDSA signatures.

Theorem 3 (CHASM is Secure Against Replay Attacks)**.** CHASM is protected against message replay attacks by classical and quantum-capable adversaries.

Proof: Under IEEE 1609.2, every BSM (denoted by M) must contain a timestamp that records when the BSM was generated (denoted by $\mathsf{t_g}$). Recall that by Theorem 2 and Corollary 2.1, neither a classical nor quantum-capable adversary can modify M and retain or produce a valid pair of digital signatures. Therefore, since $\mathsf{t_g} \in \mathsf{M}$, any modification of $\mathsf{t_g}$ to evade replay or expiration detection will result in signature verification failure and M being discarded. We thus conclude CHASM is protected from replays by classical and quantum adversaries. □

C. Discussion

Proving CHASM is EUF-CMA secure when its underlying signature algorithms are EUF-CMA secure establishes CHASM as secure against message spoofing and forgery. Importantly, we proved this against both classical and quantum-capable adversaries, fulfilling the first two design (and first two security) criteria from Section III-D. Theorem 3 further shows security against replays, meeting the remaining security criterion.

To prove the EUF-CMA security of CHASM, we assumed the underlying classical and quantum-resistant signature algorithms used to instantiate CHASM are both known to be EUF-CMA secure. In practice, this means assuming ECDSA and QRC algorithms are EUF-CMA secure, which is not necessarily the case today. For example, ECDSA is provably UF-CMA secure in the random oracle model [34], but it is not necessarily EUF-CMA secure. Similarly, it has recently been suggested by leading experts that Falcon can only be proved UF-CMA secure, while a comprehensive EUF-CMA security proof remains elusive [35]. However, our proofs still hold for a UF-CMA assumption. The key difference is that this only guarantees signatures on *new* messages cannot be forged, which is the more serious concern—EUF-CMA also prevents a new signature on an already-signed message, which is much less dangerous, as we mostly care about adversaries creating bogus and misleading messages and then signing them, which is prevented by UF-CMA. We also note that unlike Falcon, ML-DSA is EUF-CMA secure [36], underscoring the importance of CHASM's crypto-agile design.

VII. Evaluation

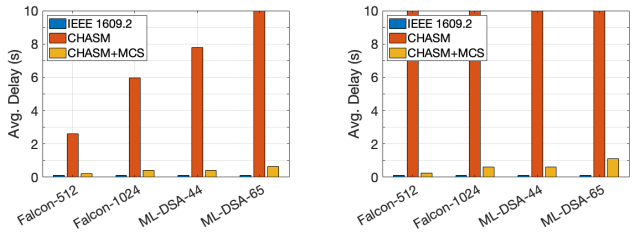
We now evaluate CHASM to study its impact on reliable and efficient V2V, considering two metrics:

Network-Induced Verification Delay. This metric is the amount of *added* delay incurred for verification of BSMs under CHASM, which is primarily seen when one vehicle receives BSMs from another vehicle whose certificate is unknown, and must then cache those BSMs to be verified after all of the certificate are received, reassembled, and verified (see Algorithm 2). We do not study other types of delay, such as the hardware execution time for verifying two digital signatures per SPDU in CHASM vs. one in current systems, because such delays are comparatively trivial—also, QRC execution time is similar to ECDSA and measured in microseconds [37].

System capacity. We define the C-V2X system capacity as the number of vehicles that can be supported in an area—roughly the ~600 m C-V2X communication range in a circle around a given vehicle—without significantly degrading communication reliability due to mutual interference of vehicles.

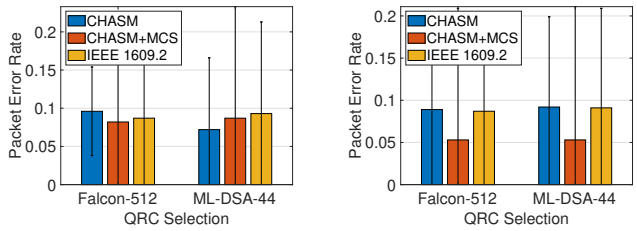
A. Experimental Setup

We use a combination of hardware experiments and simulations to evaluate CHASM, with over-the-air experiments supplemented where necessary by simulations (e.g., to study system



(a) Low Density (PER = 0.01). (b) High Density (PER = 0.1).

Fig. 6. Network delay for BSM verification. Results are shown for CHASM with different QRC in both low-density (a) and high-density (b) scenarios.



(a) Low Density. (b) High Density.

Fig. 7. Rural (a) and urban (b) results show CHASM meets the ≤10% PER requirement. Results are mean across all vehicles over several minutes.

capacity, which involves scenarios with far more vehicles than we can feasibly account for with hardware devices).

Over-the-Air Experiments. We use USRP X310 software-defined radios (SDRs) with UBX daughterboards and internal GPS modules. We also use a USRP B210 with an internal GPS module, and synthetic GPS signals are supplied to all SDRs using a LimeSDR and `gps-sdr-sim` [38]. Each USRP can send and receive C-V2X signals using software we develop by extending the open-source srsRAN project [39] to support sidelink. For the application layer and security services, we implement CHASM as well as the relevant IEEE and SAE protocols [9], [24] for exchanging properly formatted SPDUs. For cryptography, we use OpenSSL for ECDSA (with the P.256 elliptic curve specified by IEEE 1609.2), and we use `liboqs` [40] for NIST-compliant QRC.

Simulation Software. We extend the Artery+C [41] C-V2X simulator to support CHASM and emulate QRC primitives (i.e., properly adjusting packet sizes to match actual values).

B. Results — Verification Delay

We use our hardware testbed to perform over-the-air measurements with real signals. We position our USRP X310s approximately 3 m apart, with each X310 playing the role of one vehicle, and we configure both to run CHASM. To emulate the presence of other vehicles, we use our USRP B210 to transmit a large volume of traffic from "all other vehicles," varying its transmit gain between transmissions to accurately approximate the received power for different vehicles at varying distance from the two devices under study. Over 1000 iterations of a Monte Carlo experiment, we begin transmitting SPDUs between the X310s and track the average time between when a BSM is received and when it can be verified using the sender's certificate (which is also sent over the air), yielding the average *verification delay*. We replicate this experiment for three configurations (IEEE 1609.2, CHASM, and CHASM with our MCS enhancement) with four QRC algorithms (Falcon-512, Falcon-1024, ML-DSA-44, and ML-DSA-65). We also consider two different roadway scenarios, one with a low density of traffic (30 vehicles) and one with a higher density (100 vehicles), in which the measured packet error rate (PER) values are approximately 0.01 and 0.1, respectively.

Our results (Fig. 6) illustrate the importance of using CHASM *and* our MCS enhancement in a complete, cross-layer solution. In both the low- and high-density roadways, the average delay incurred when using CHASM alone is very significant (often in excess of 10 seconds), showing that certificate fragmentation alone is not enough. However, when we combine CHASM with our MCS optimization, the delay is far less, approximating current delays under IEEE 1609.2 across both scenarios and irrespective of the QRC algorithm that is used. Importantly, this not only demonstrates the practicality of our CHASM+MCS approach, it also proves that our protocol is *crypto-agile*, supporting a range of NIST-approved QRC without significant performance differences across algorithms.

C. Results — System Capacity

To demonstrate that CHASM can handle enough vehicles for current and future roadways, we run simulations considering two scenarios representing rural (low-density) and urban (high-density) roadways. We select average vehicle speeds (resp. 95 and 45 km/hr) and densities (resp. 10 and 100 vehicles/km) based on historical government traffic data [42]. All vehicles transmit at 23 dBm on 5.92 GHz in a 20 MHz channel. Channel effects are applied based on standard 3GPP models [6]. We again consider multiple QRC choices in each scenario (only two are shown, for brevity), and we evaluate the same three configurations as before—current IEEE 1609.2, CHASM with varying QRC, and CHASM with our MCS enhancement.

Our results, shown in Fig. 7, conclusively demonstrate that CHASM—both alone and with our MCS enhancement—does not unacceptably impact PER. Indeed, not only is PER with CHASM similar to PER under current IEEE 1609.2 systems, we find that for all QRC choices in both scenarios, PER meets the required 10% cap set by SAE J3161 [9]. Moreover, we note that in the high-density scenario we are considering 100 vehicles/km of roadway, which is actually an overestimate when compared with rush-hour traffic data on multi-lane highways in the U.S. [42]. Therefore, we conclude that CHASM is fully capable of supporting the requisite number of vehicles expected on today's roads, even up to a connected vehicle ratio approaching 100% vs. legacy (non-connected) vehicles.

VIII. Related Work

To the best of our knowledge, we are the first to develop a practical technique for quantum-resistant BSM authentication in C-V2X. Only one prior work attempts to solve this problem—in that work [14], the authors developed quantum-resistant security *only for certificates*, leaving BSMs protected with just ECDSA signatures. This makes their approach insecure once quantum computers can break ECDSA

in less than 5 minutes; in contrast, CHASM provides *complete* quantum-resistant security against near- and long-term threats. Also, [14] focuses on a different (and now-outdated) V2V protocol with less severe constraints than C-V2X. The few other works on quantum resistance for CVs are surveys that do not offer novel solutions (e.g., [43]), are not backwards-compatible [44], or make unrealistic assumptions (e.g., [45] assumes a payload limit of $12,000$ bytes vs. a realistic $\sim$800).

Outside of CVs, many works have proposed hybrid authentication protocols to support QRC, including in embedded devices [46] and other constrained environments. However, such works are designed for very different types of communication—for example, where wired connections are available or bidirectional techniques like key negotiation can be employed [47]. In general, such approaches cannot be adapted to the wireless, broadcast-only V2V environment. Moreover, most such works focus on confidentiality or access control (e.g., [48]) as opposed to our focus on authentication.

IX. Conclusions and Future Work

We introduced CHASM, which is designed to enable an immediate, standards-compliant transition to quantum-resistant authentication in current and future C-V2X. We formally proved CHASM is secure against classical and quantum-capable adversaries. We further demonstrated its practicality by showing that CHASM can support at least 100 vehicles/km with high reliability, which exceeds the expected density of connected vehicles on the roads for the foreseeable future. Critically, we also established that adopting our MCS enhancement alongside CHASM avoids excessive BSM verification delays, confirming our insight that a *cross-layer approach* is needed to fully support QRC in real-world scenarios. In future work, we will expand our scope for threat models and proofs, as well as expand our experiments to cover large-scale real-world tests.

References

[1] National Highway Traffic Safety Administration, "NHTSA estimates 39,345 traffic fatalities in 2024," Apr. 2025, https://bit.ly/4mKUsHS.

[2] European Commission, "2024 sees 3% drop in EU road fatalities, yet progress remains slow," Mar. 2025, https://tinyurl.com/upfwbrcz.

[3] D. Moody, R. Perlner, A. Regenscheid, A. Robinson, and D. Cooper, "Transition to post-quantum cryptography standards," National Institute of Standards and Technology (NIST), Internal Report 8547, Nov. 2024.

[4] M. Mosca and M. Piani, "Quantum threat timeline report 2024," Dec. 2024. [Online]. Available: https://tinyurl.com/33puur85

[5] 3GPP, *Release 14: Release Description*, 3GPP TR 21.914, 2018.

[6] ——, *Summary of Rel-16 Work Items*, 3GPP TR 21.916, Jul. 2020.

[7] ——, "Releases," 2025, https://bit.ly/4mHSdoM.

[8] *Physical layer procedures for data*, 3GPP TS 38.214, Jul. 2022.

[9] *On-Board System Requirements for LTE-V2X V2V Safety Communications*, SAE Std. J3161/1, Mar. 2022.

[10] C. Tjhai *et al.*, *Multiple Key Exchanges in the Internet Key Exchange Protocol Version 2 (IKEv2)*, IETF RFC 9370.

[11] M. Singh and D. Mishra, "Post-quantum secure authenticated key agreement protocol for wireless sensor networks," *Telecommunications Systems*, vol. 84, pp. 101–113, Jul. 2023.

[12] D. Stebila, S. Fluhrer, and S. Gueron, *Hybrid key exchange in TLS 1.3*, Internet-Draft, Apr. 2024. [Online]. Available: https://bit.ly/4dsaBwI

[13] D. D. Tran, K. Ogata, S. Escobar, S. Akleylek, and A. Otmani, "Formal analysis of post-quantum hybrid key exchange SSH transport layer protocol," *IEEE Access*, vol. 12, pp. 1672–1687, Dec. 2023.

[14] G. Twardokus, N. Bindel, H. Rahbari, and S. McCarthy, "When cryptography needs a hand: Practical post-quantum authentication for V2V communications," in *Proc. NDSS Symposium*, San Diego, CA, Feb. 2024.

[15] National Automobile Dealers Association, "US consumers keep vehicles for a record 12.5 years on average - S&P (Reuters)," 2023, Available: https://tinyurl.com/3jc224u8. Accessed: Jan. 21, 2025.

[16] Bureau of Transportation Statistics, "Average age of automobiles and trucks in operation in the United States," 2022, Accessed: June 21, 2023. [Online]. Available: https://bit.ly/3PosWR3

[17] US Government Accountability Office, "Vehicle safety: Opportunities to improve repair rates for recalled vehicles," 2024, GAO-24-106356. Accessed: Jan. 21, 2025. https://www.gao.gov/assets/d24106356.pdf.

[18] 5GAA, "List of C-V2X devices," Apr. 2024, Accessed: Apr. 8, 2025. [Online]. Available: https://bit.ly/45HlEkH

[19] *Architecture enhancements for 5G System to support V2X services*, 3GPP Technical Specification 23.287, 2021.

[20] NIST Computer Security Resource Center, "Post-quantum cryptography: FAQs," jan 2020. [Online]. Available: https://tinyurl.com/2cmar3dm

[21] Federal Office for Information Security (BSI), "Quantum-safe cryptography: fundamentals, current developments and recommendations," Tech. Rep., 2021. [Online]. Available: https://tinyurl.com/2c5b3mkz

[22] Federal Communications Commission, "FCC national broadband map," Dec. 2024. [Online]. Available: https://bit.ly/4fw3dCC

[23] *5G; NR: User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone*, 3GPP TS 38.101 V16.1.1, Jul. 2020.

[24] *Wireless Access in Vehicular Environments—Security Services for Application and Management Messages*, IEEE Std. 1609.2, 2022.

[25] USDOT, "Interoperable connectivity/spectrum," 2025, Accessed: Aug. 28. 2025. [Online]. Available: https://tinyurl.com/2vpf87k9

[26] *On-Board System Requirements for V2V Safety Communications*, SAE International Surface Vehicle Standard J2945/1, Apr. 2020.

[27] Federal Communications Commission, "FCC Docket No. 19-138," Nov. 2020. [Online]. Available: https://bit.ly/4c9aCop

[28] R. Wu, "C-V2X automotive tech brings enhanced safety and efficiency to China's roads," Qualcomm, Mar. 2021, Accessed: Aug. 1, 2024. [Online]. Available: https://bit.ly/3ydjFXA

[29] B. Aizpurua, P. Bermejo, J. E. Martínez, and R. Orus, "Hacking cryptographic protocols with advanced variational quantum attacks," *ACM Trans. Quantum Computing*, vol. 6, no. 2, pp. 141–124, Mar. 2025.

[30] M. Sobhani, Y. Chai, T. Hartung, and K. Jansen, "Variational quantum eigensolver approach to prime factorization on IBM's noisy intermediate scale quantum computer," *Phys. Rev. A*, vol. 111, no. 4, Apr. 2025.

[31] P. W. Shor, "Algorithms for quantum computation: Discrete logarithms and factoring," in *Proc. Annual Symposium on Foundations of Computer Science,*, Santa Fe, NM, Nov. 1994, pp. 124–134.

[32] S. Balumuri, E. Eaton, and P. Lamontagne, "Quantum-safe public key blinding from MPC-in-the-head signature schemes," in *Proc. Int. Conf. Appl. Crypto. Netw. Security*, Munich, Germany, Jun. 2025, pp. 369–397.

[33] A. Goldsmith, *Wireless Commun.* Cambridge University Press, 2005.

[34] A. Boldyreva, V. Kumar, and J. Sun, "Provable security analysis of butterfly key mechanism protocol in IEEE 1609.2.1 standard," in *Proc. Priv. Enhanc. Tech. Symp. (PETS)*, Bristol, UK, Jul. 2024, pp. 565–582.

[35] P. Gajland, J. Jannec, and E. Kiltz, "A closer look at falcon," Feb. 2025. [Online]. Available: https://eprint.iacr.org/2024/1769

[36] U.S. Dept. of Commerce, "Module-lattice-based digital signature standard," Federal Information Processing Standard (FIPS) 204, 2024.

[37] D. T. Nguyen and K. Gaj, "Fast Falcon signature generation and verification using ARMv8 NEON instructions," in *Proc. AFRICACRYPT 2023*, Sousse, Tunisia, Jul. 2023, pp. 417–441.

[38] T. Ebinuma, "GPS-SDR-SIM," 2018, Accessed: Apr. 20, 2020. [Online]. Available: https://github.com/osqzss/gps-sdr-sim

[39] I. Gomez-Miguelez, A. Garcia-Saavedra, P. Sutton, P. Serrano, C. Cano, and D. Leith, "SrsLTE: An open-source platform for LTE evolution and experimentation," in *WINTECH 2016*, New York City, USA, Oct. 2016.

[40] N. Allen *et al.*, "liboqs," 2025, https://tinyurl.com/muv8x3ej.

[41] A. Hegde and A. Festag, "Artery-C: An OMNeT++ based discrete event simulation framework for cellular V2X," in *Proc. MSWiM 2020*, Alicante, Spain, Nov. 2020, pp. 47–51.

[42] G. Twardokus and H. Rahbari, "Towards protecting 5G sidelink scheduling in C-V2X against intelligent DoS attacks," *IEEE Transactions on Wireless Communication*, vol. 22, no. 11, pp. 7273–7286, Nov. 2023.

[43] N. Lohmiller, S. Kaniewski, M. Menth, and T. Heer, "A survey of post-quantum cryptography migration in vehicles," *IEEE Access*, vol. 13, pp. 10 160–10 176, Jan. 2025.

[44] Y. Gong and B.-J. Hu, "A quantum-resistant key management scheme using blockchain in c-v2x," *IEEE Trans. on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 16 831–16 844, Nov. 2024.

[45] Y. Kim and S. Seo, "Signature split method for a PQC-DSA compliant with V2V communication standards," *Appl. Sci.*, vol. 13, May 2023.

[46] G. Tasopoulos, J. Li, A. P. Fournaris, R. K. Zhao, A. Sakzad, and R. Steinfeld, "Performance evaluation of post-quantum TLS 1.3 on resource-constrained embedded systems," in *Proc. Int. Conf. Info. Security Practice and Exp.*, Taipei, Taiwan, Nov. 2022, pp. 432–451.

[47] J. Yao, K. Matusiewicz, and V. Zimmer, "Post quantum design in SPDM for device authentication and key establishment," *Cryptography*, vol. 6, no. 4, p. 48, 2022.

[48] V. Q. Ulitzsch, S. Park, S. Marzougui, and J.-P. Seifert, "A post-quantum secure subscription concealed identifier for 6G," in *Proc. ACM Conf. Security and Privacy in Wireless and Mobile Netw. (WiSec)*, San Antonio, TX, 2022, pp. 157–168.